THE CHINESE ECONOMY

A JOURNEY INTO HISTORY AND ECONOMICS

Lecture 2: How do we know? The Great Famine

WHERE WE STOPPED: THE GREAT LEAP FORWARD, 1958-60

- **Goal: industrialise fast**
  - Everything at once, regardless of real resource constraints: "a vision, rather than a plan"
- **Organisation**: communes; bonuses and free markets abolished; decisions decentralised; "walking on two legs"
- **Two fateful decisions**: less labour and land for farming; more grain taken by the state
- **Why nobody braked**: inflated reports — the statistical system, the leaders' "eyes and ears", collapsed

WHAT HAPPENED, 1958-62

- **1958**: the Great Leap; communes and communal kitchens; labour and land pulled out of farming; harvests reported far above reality
- **Summer 1959, Lushan**: a party conference meant to correct course. Mao turns on the critics (Peng Dehuai, minister of defence); a new Anti-Rightist campaign; **grain procurement reaches new heights**
- **Harvests fall**: grain per person 292 kg (1958), 246 (1959), 214 (1960)
- **1960**: full famine, rural, worst inland (Sichuan, Anhui, Guizhou); the coastal cities stay supplied. Soviet advisers withdrawn in the summer
- **Excess deaths**: estimates from **16.5 to 45 million**
- **After the Leap**: retreat; scarce foreign exchange goes to **grain imports**

Naughton (2007), pp. 69-72, 240, 379; Meng, Qian, Yared (2015), Table 1 and fn. 1

"THREE PARTS NATURE, SEVEN PARTS MAN"

- **Liu Shaoqi**, President of the PRC, early 1962 (Seven Thousand Cadres Conference): the famine was **30% natural disaster, 70% human error** (三分天灾，七分人祸)
- **The Party, 1981**: mainly the mistakes of the Great Leap and of the campaign against "right deviation", plus natural disasters and the split with the Soviet Union
- **Meng, Qian and Yared, 2015**: the way grain was procured accounts for about **32-43%** of famine deaths

What is the difference between these numbers?

THE PUZZLE

In 1959-61, average food availability in the Chinese countryside was enough to feed the population.

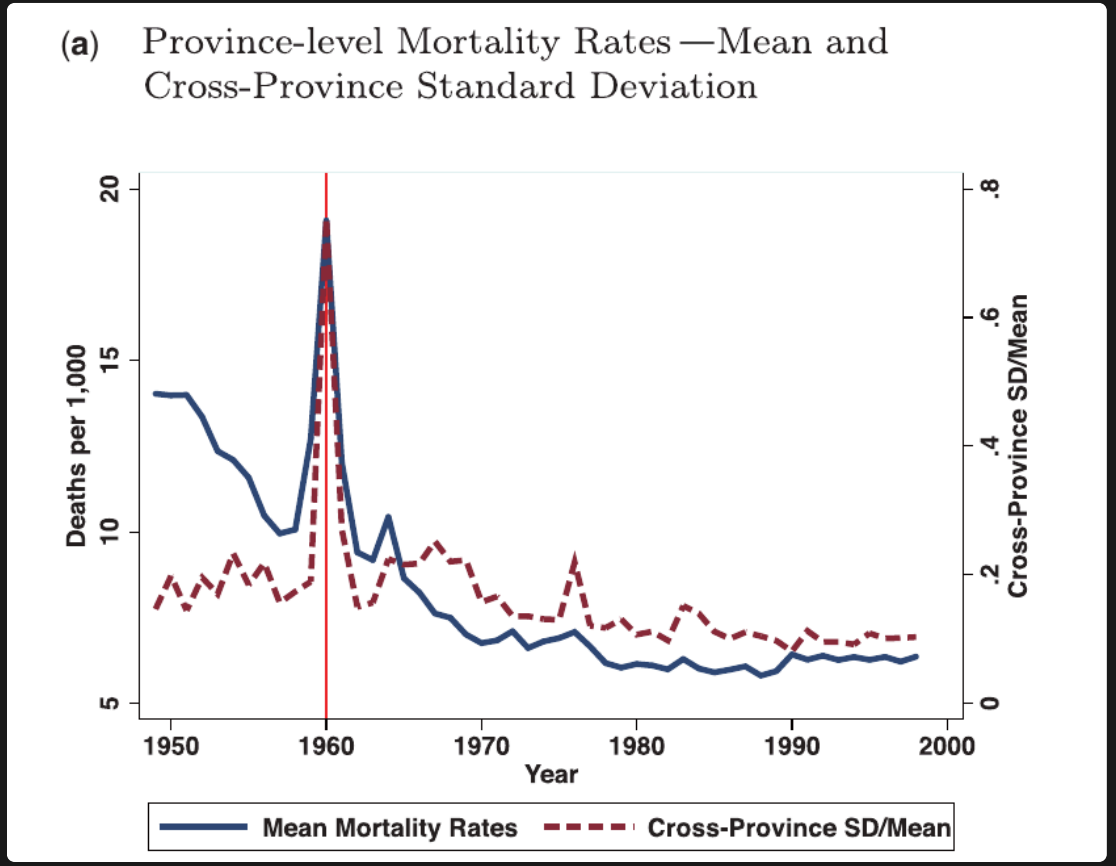
So why did 16.5 to 45 million people die?

- After the state took its share, the rural average in 1959 was about **2,300 kcal a day**; the survival benchmark is about **800**
- Production per person fell about 15% in 1959 — the largest fall of the period — but stayed far above that line

Meng, Qian, Yared (2015), *Review of Economic Studies* — 19 provinces, 77% of the population

IF FOOD WAS UNEQUAL, DEATHS SHOULD BE UNEQUAL

- The average was enough. So the food must have been unequal. **If that is true, what else must we see?**
- In 1960 the national average death rate jumps from about 10 to about **19 per 1,000**
- And so does the *spread* between provinces (dashed line): the same national shock, very different local outcomes
- That is what "uneven" means: not just more deaths, but more different deaths



Meng, Qian, Yared (2015), Fig. 1a

CAUSES WE TAKE FOR GRANTED

- "Smoking causes lung cancer"
- "Drinking causes car accidents"
- "One more year of school raises your wage"
- "Fixed grain quotas killed millions of people"

We say these every day. How do we know any of them?

WHAT DOES "X CAUSES Y" MEAN? THREE ANSWERS

	X causes Y if…	Smoking and cancer	The famine
Granger prediction	X comes first and helps **predict** Y	Smokers today are more likely to fall ill later	Grain output this year predicts deaths next year
Rubin counterfactual	without X, in a world we do not see, Y **would have been different**	Had the same person never smoked, no cancer	With a lower quota, the same province in the same year would have lost fewer people
Pearl graph	**acting on X from outside** changes Y; a graph says what "from outside" means	Make people stop, and cancer falls	Lower the quota from outside, leave the rest as it was

Today we use Rubin, and draw it the Pearl way

Clip shown in class (not online)

Sliding Doors (Peter Howitt, 1998)

THE FUNDAMENTAL PROBLEM OF CAUSAL INFERENCE

- Helen catches the train: **Y(1)**. Helen misses it: **Y(0)**. The effect of the train is the difference
- The film shows both lives. Real life shows **one**
- You, with 5 euros tonight, and you without them: same problem
- **Either Y(1) or Y(0) is observed, never both.** The missing one is the **counterfactual**
- The famine: the same province, in the same year, with a lower quota. Nobody will ever see it

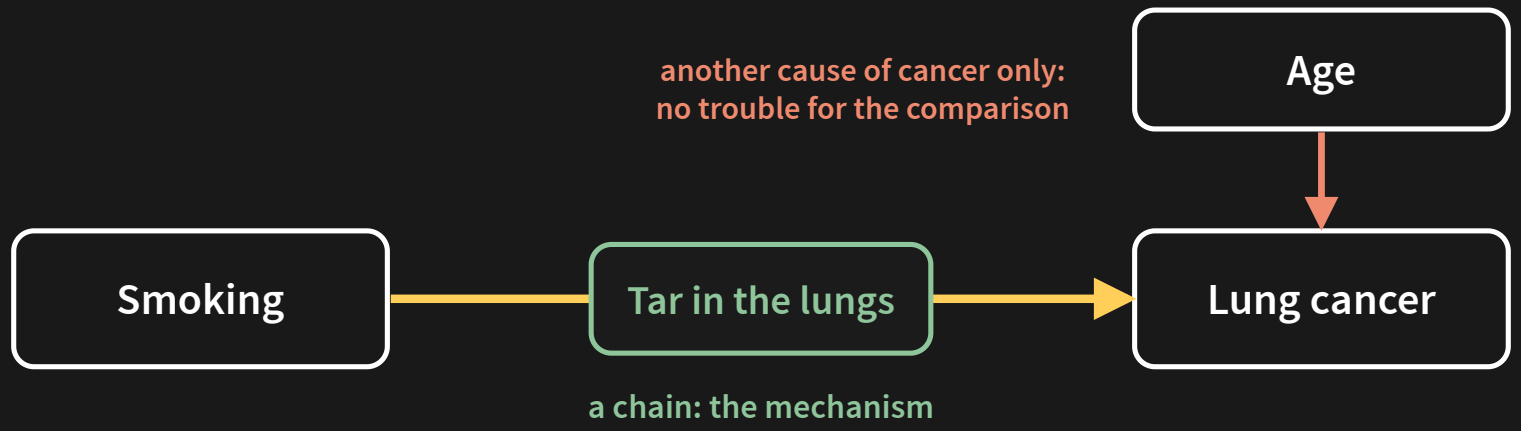
THE SOLUTION: RANDOMISATION

- **Give the 5 euros by lottery**: half of you win, half do not
- **Why it works**: winning has nothing to do with you: hunger, wallet, mood, things we cannot even see. With many people, the two groups end up **alike on average** in all of them
- So the losers' average stands in for the winners' missing Y(0): **the gap between the two averages is the average effect**
- If instead **I choose** (the hungriest, the front row), the groups differ before any money changes hands: the gap mixes the effect with the choice
- Effects differ from person to person: the average hides who gains more and who gains less

WE CANNOT RANDOMISE A FAMINE

- Nobody ran a lottery. **History** decided which provinces had high quotas, and history is not random
- So the question becomes: **can we tie the deaths to a relevant factor that was allocated *as if* at random?**
- The paper's candidate: grain **expected from the weather and the land**. Nobody chose those
- Before using it: what can go wrong?

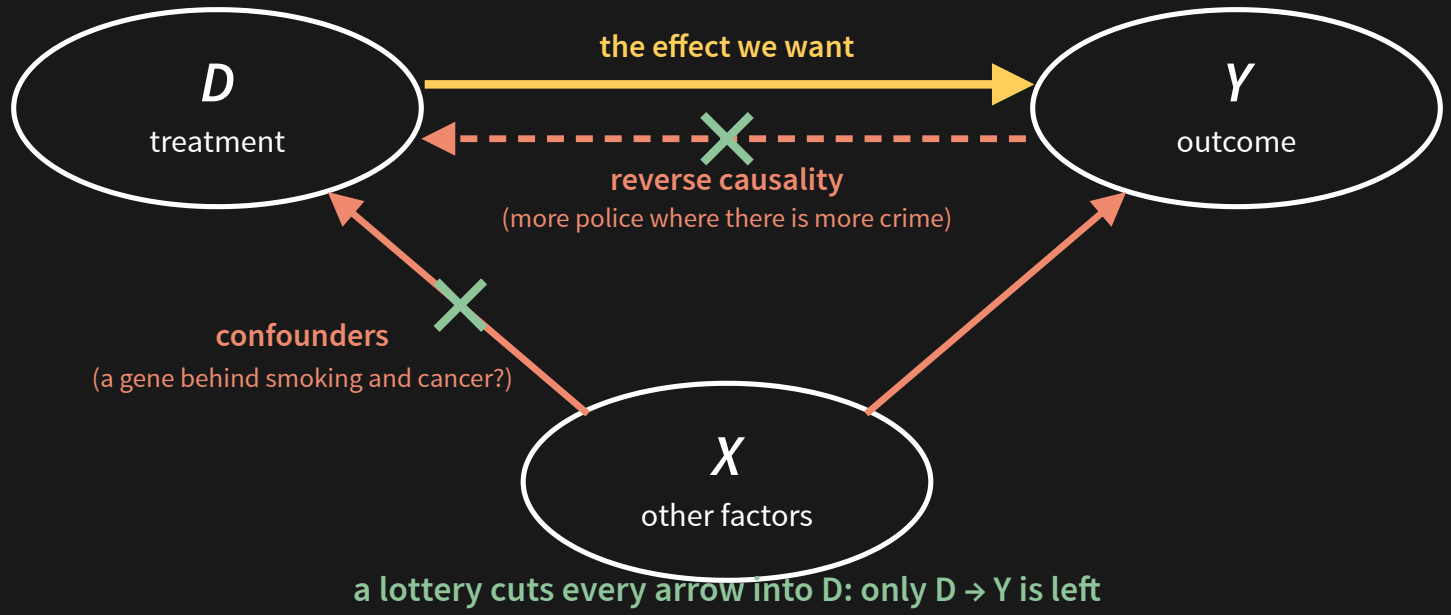
A CAUSAL GRAPH, THE SIMPLEST



What is not drawn, we assume is not there

Trouble starts only with arrows that reach **both** the cause and the effect. Next slide

THE EMPIRICAL CHALLENGE



WHERE DID MORE PEOPLE DIE? COMPARE AVERAGES

Split the provinces in two: **high** and **low** expected grain per head. In each group, take the **average death rate** — in normal years and in famine years.

	Normal years	**Famine years**
High expected grain	about the same	**higher**
Low expected grain	about the same	lower

What would you expect? More food, fewer deaths?

An average *inside a group* is a **conditional average**. The finding is a **difference of differences**: the gap between the groups appears only in the famine years — and it goes the "wrong" way.

READING THE TABLE: A DIFFERENCE IN DIFFERENCES

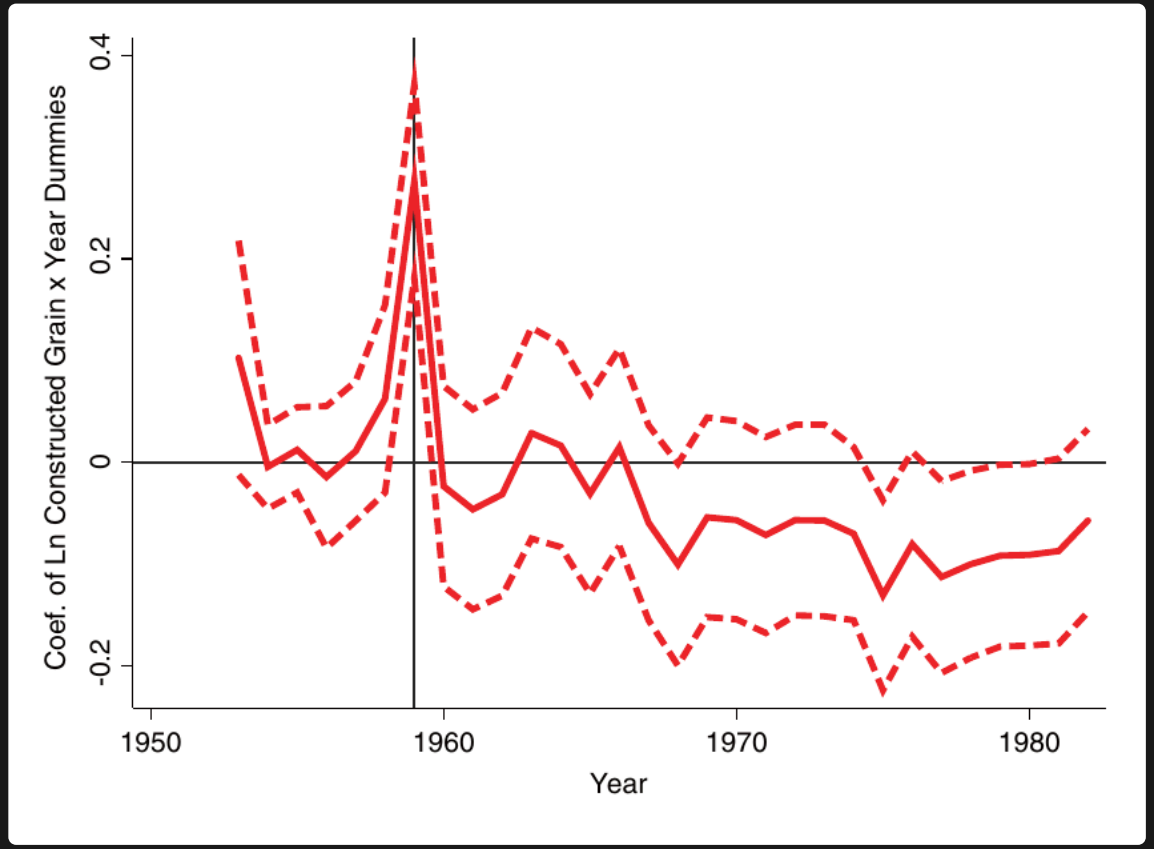
Deaths per 1,000 people a year. **Illustrative numbers**, with the sign and rough size of the paper's estimates

	Normal years	Famine years	Change
High expected grain	10	22	+12
Low expected grain	11	18	+7
Gap, high minus low	−1	+4	**+5**

- Down the last column (12 − 7) or across the last row (4 − (−1)): **the same +5**. That is the **difference in differences**
- How to say it: in the famine years, deaths in high-grain provinces rose by **5 more per 1,000** than in low-grain provinces, compared with normal years
- Points or percent? High-grain deaths rose **120%**, low-grain **64%**: a gap of **56 percentage points** — not "56%"

THE SAME TABLE, YEAR BY YEAR

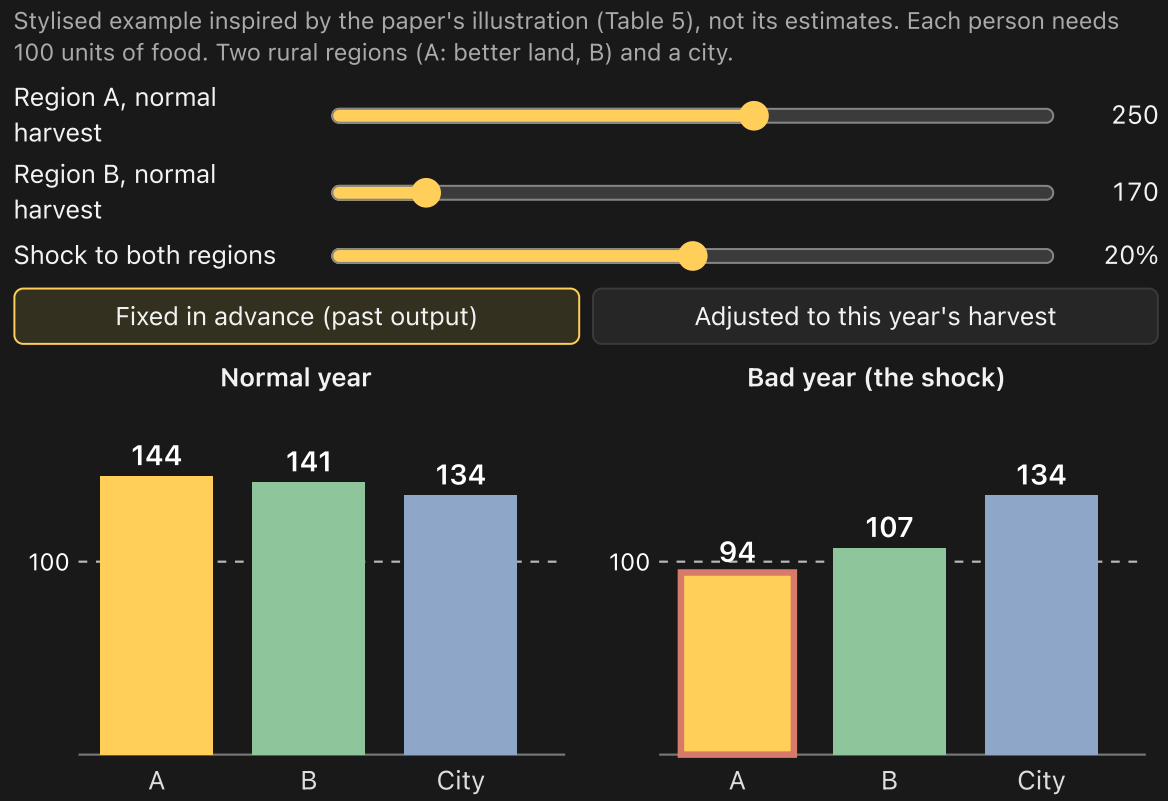
- We would expect more deaths where there was *less* food
- Across provinces, the paper finds the opposite: **more deaths where predicted grain output per head was higher** — only in the famine years
- How to read it: each point is a year — how much higher mortality is where expected grain is higher. Zero means no gap. Dashed lines: the range we cannot tell apart from noise
- About zero in normal years; clearly positive in 1958-59



Meng, Qian, Yared (2015), Fig. 3

HOW THE STATE COLLECTED GRAIN: A TOY ECONOMY

- Quotas were set **in advance**, from past verified output, top-down (province, prefecture, county)
- Each collective delivered its quota **even if the harvest fell short**
- Regions with higher output faced a **higher share** taken
- Try it: with the default numbers, after the shock region A (better land) is left with **less** than region B



HOW DO WE MEASURE THIS WITH BAD DATA?

- **Both** numbers we just compared, grain **and deaths**, are the state's own statistics, unreliable in these years
- **Proxy 1**: output *predicted* from weather, geography and rural population, not the reported one
- **Proxy 2**: the missing children. In the 1990 census, how large is each birth cohort, in 511 counties?
- A small cohort born in 1959-61 = more deaths and fewer births. And the counties with better land lose the most cohorts (right)



Fig. 1b: cohort size, by birth year

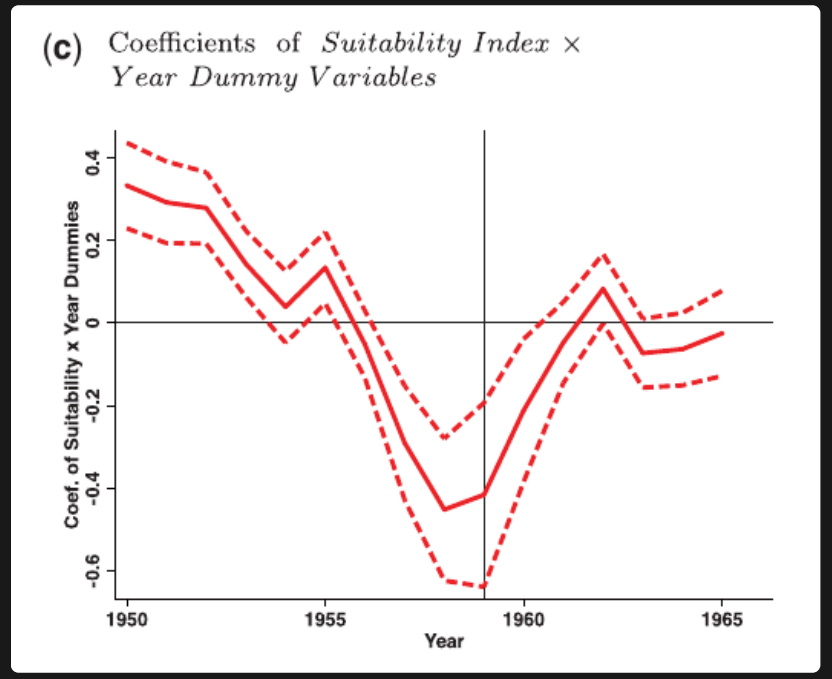
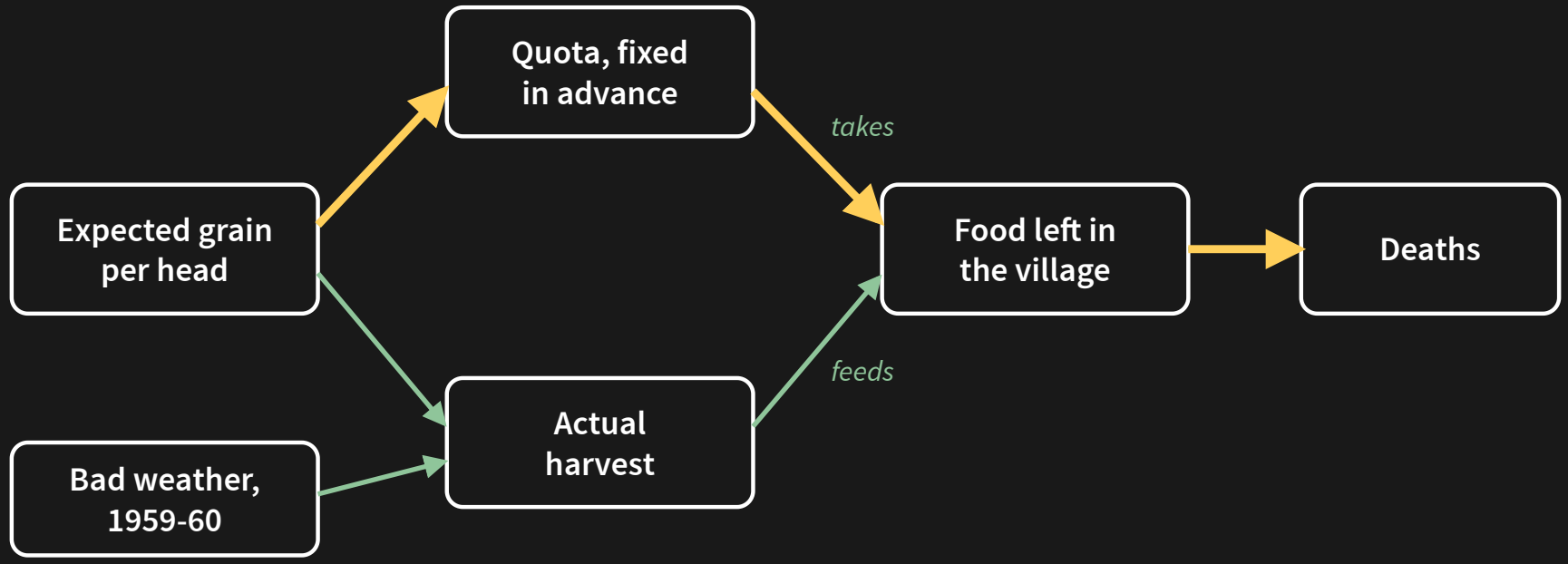


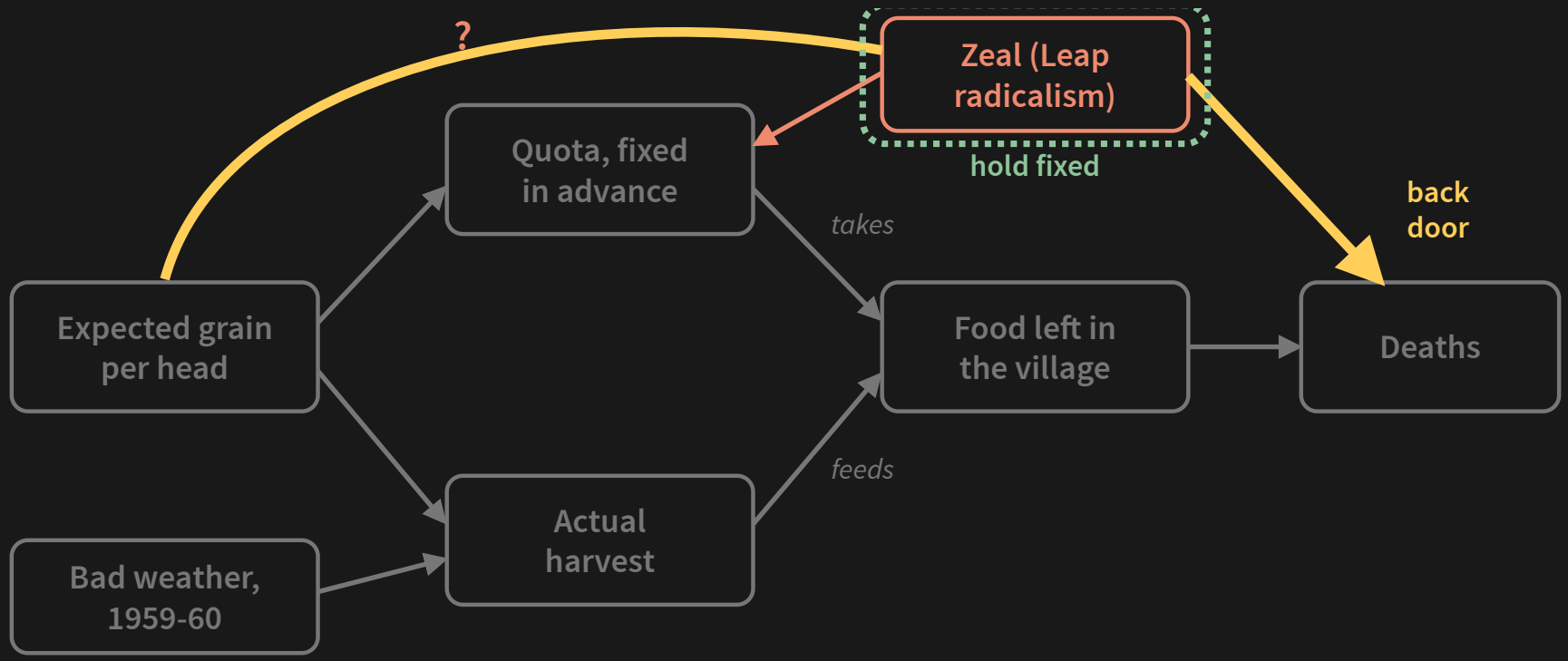
Fig. 4c: land suitability x birth year

THE FAMINE AS A CAUSAL GRAPH



Arrows say what causes what. No loops. **What is not drawn, we assume is not there.**

A RIVAL STORY: GREAT LEAP ZEAL



- Zeal kills on its own (communal kitchens, labour sent to steel) and pushes quotas up
- Were provinces with more grain also more zealous? If so, the gap could be zeal, not grain
- That path, grain ← zeal → deaths, is a **back door**: a **confounder**
- To close it: compare high and low grain **at the same level of zeal**

CLOSING THE BACK DOOR: COMPARE AT THE SAME ZEAL

Less zealous provinces

	Rise in deaths, famine vs normal
High grain	bigger
Low grain	smaller

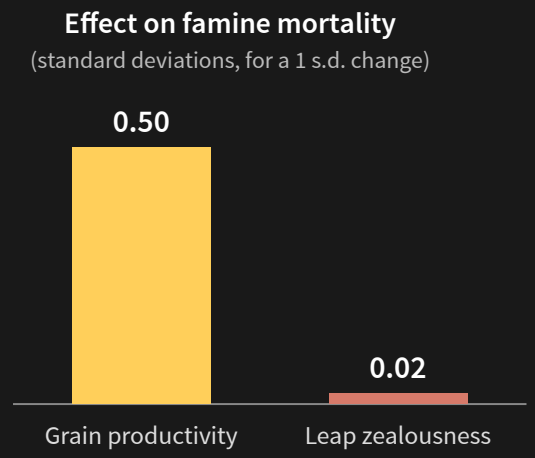
More zealous provinces

	Rise in deaths, famine vs normal
High grain	bigger
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- The gap **survives** when zeal is held fixed
- Which way could zeal have pushed? Provinces with more grain were, if anything, *less* zealous. Leaving zeal out would hide part of the gap, not create it
- That is the idea of "controlling for" something. The paper does it with a regression; the logic is these two tables

IS IT JUST GREAT LEAP ZEAL? AND HOW BIG IS IT?

- Robustness of equation (1). Alternative: Great Leap **zealousness** (sown-area cuts, steel output, communal dining, purges, late liberation), urban bias, misreported output
- First check, in 17 provinces: were more productive provinces more zealous? If anything, *less*
- Add the zeal measure (their first principal component) and its interaction with the famine dummy to equation (1): the result barely changes
- Back-of-the-envelope: the procurement mechanism accounts for roughly **one third to 43%** of famine mortality — which leaves room for the Leap's policies and for politics



NINETEEN PROVINCES, TOO MANY QUESTIONS



- Every new thing we hold fixed splits the groups again. Soon we compare **one province with one province**
- Two ways out: **more units** (the paper goes down to 511 counties) or **a model** that fills the empty cells by assuming a shape: **regression**. Next lecture

THE IDEAL EXPERIMENT, AND THE REAL ONE

Ideal experiment	What history gave	What the paper does
Identical provinces	Provinces differ: land, cities, zeal	Compares the gap between provinces in famine years with the same gap in normal years; holds zeal fixed
A lottery decides who gets a high quota	Quotas follow past output	Expected grain from weather and land: things nobody chose
True harvests measured	Harvests over-reported	Rebuilds output from weather, land, population
Deaths counted	Nobody counted	Missing children in the 1990 census
People stay where the treatment is	Peasants could not leave (hukou)	Nothing to do: here history helps

For every gap between the two columns: **which way would it push the result?**

BACK TO LIU SHAOQI: 70% AND 32-43%

Liu Shaoqi, 1962

- A verdict on one country and one famine
- No comparison: nature and man are mixed in the same total
- Said by a leader, at a party meeting
- Nothing that could have proved it wrong

Meng, Qian, Yared, 2015

- One mechanism, not the whole famine
- A comparison of averages across provinces and years
- Measures built from data nobody could manipulate
- A rival story checked; the rest left open
- It could have come out wrong

Not "one is right, one is wrong". One can be checked.

QUESTIONS, FOR NEXT WEEK



"Eliminate the four pests!" (1958), and a boy with the day's catch

1. A silent spring. In 1958 the Four Pests campaign killed sparrows all over China; insects that sparrows used to eat then multiplied in the fields. **Add the campaign to the causal graph**: which boxes, which arrows?

2. Migration. Peasants could not leave their villages. Suppose they could have. **Add migration to the causal graph**: which boxes, which arrows? Would the effect of the quotas on deaths go **up or down**?

3. What was the least clear point in Lecture 1? And today, in Lecture 2?

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